

Lab program-31:

31. Construct a C program to simulate the First in First Out paging technique of memory management.

Code:

```
#include <stdio.h>

#define MAX_FRAMES 3

void printFrames(int frames[], int n)
{
    for (int i = 0; i < n; i++)
    {
        if (frames[i] == -1)
        {
            printf(" - ");
        }
        else
        {
            printf(" %d ", frames[i]);
        }
    }

    printf("\n");
}

int main()
{
    int referenceString[] =
        {7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2};

    int n = sizeof(referenceString) / sizeof(referenceString[0]);

    int frames[MAX_FRAMES];
    int framePointer = 0;

    // Initialize frames
    for (int i = 0; i < MAX_FRAMES; i++)
    {
        frames[i] = -1;
    }

    printf("Reference String: ");

    for (int i = 0; i < n; i++)
    {
```

```

        printf("%d ", referenceString[i]);
    }

    printf("\n\n");

    printf("Page Replacement Order:\n");

    for (int i = 0; i < n; i++)
    {
        int page = referenceString[i];
        int pageFound = 0;

        // Check if page is already in memory
        for (int j = 0; j < MAX_FRAMES; j++)
        {
            if (frames[j] == page)
            {
                pageFound = 1;
                break;
            }
        }

        // Page fault
        if (!pageFound)
        {
            printf("Page %d -> ", page);

            frames[framePointer] = page;

            framePointer = (framePointer + 1) % MAX_FRAMES;

            printFrames(frames, MAX_FRAMES);
        }
    }

    return 0;
}

```

OUTPUT:

```
PS C:\Users\SMILE\Documents\OS\Lab programs> ./lab_31.exe
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2

Page Replacement Order:
Page 7 -> 7 - -
Page 0 -> 7 0 -
Page 1 -> 7 0 1
Page 2 -> 2 0 1
Page 3 -> 2 3 1
Page 0 -> 2 3 0
Page 4 -> 4 3 0
Page 2 -> 4 2 0
Page 3 -> 4 2 3
Page 0 -> 0 2 3
```